



Bangalore Institute Of Technology
K R Road, V V Pura
Bengaluru-560004
www.bit-Bangalore.edu.in



CONTROL SYSTEMS(BEC403)

CONTROL SYSTEM ASSIGNMENT PROJECT INITIAL REVIEW PRESENTATION ON

“Design and Implementation of Smart Traffic Control Management System”

Presented by

USN NUMBER	FULL NAME
1BI23ET020	JEEVAN R
1BI23ET023	KRITHIKSHA S
1BI23ET027	MANASA S K
1BI23ET032	MISHRA AMBUJ SANJAY

Under the Guidance of

Dr. Girish Kumar N G

B.E, M.Sc (UK), M.Sc(Engg) by Research, Ph.D

Assistant Professor

Dept. of ETE

Bangalore Institute of Technology

Bangalore-560004

Department of Electronics and Telecommunication Engineering
Bangalore Institute of Technology, Bangalore-560004

CONTENT

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- Literature Review – papers, textbooks, official or ethical webpage referred for choosing the project **(1 Slides)**
- Problem Statement – the problem you stated after doing literatures survey**(1 Slide)**
- Methodology planned for your Work (Details Of Hardware And Software Used) **(1 Slide)**
- System Model **(1 Slides)**
- Expected Results **(2 Slides)**
- Applications of your Work**(1 Slide)**
- References- list of papers, textbooks, official or ethical webpages reffered**(1 Slide)**

Introduction

Smart Traffic Control Management System

- Traffic congestion is a major issue in urban areas, leading to delays, fuel wastage, and pollution.
- Traditional systems use fixed signal timers and manual control, which are inefficient during peak hours.
- Smart Traffic Management integrates IoT, AI, and real-time data analytics to dynamically optimize signal timings.
- Benefits: Reduced congestion, improved safety, and efficient urban mobility.

Emergency Vehicle Priority in High-Density Traffic

- **Emergency vehicle detection** using **smartphone GPS** to prioritize ambulance passage.
- **Real-time monitoring** and control using an **ESP32 microcontroller**.
- Benefits: Faster emergency response, reduced delays, and improved patient survival rates.



Literature Review

1. Papers on Smart Traffic Control Management Systems

"Smart Traffic Control System Using IoT" by P. R. Srinivas, S. Thirumalai, and K. R. Pradeep

- **Summary:** This paper explores the use of IoT-based sensors and devices to monitor traffic in real-time. It discusses how traffic management can be improved through adaptive control systems that change the traffic light timings based on the real-time density of traffic.
- **Link:** [IEEE Xplore: Smart Traffic Control](#)

2. Ethical and Official Guidelines for Implementing Smart Traffic Systems

"National Highway Traffic Safety Administration (NHTSA)" - Smart Infrastructure Guidelines

- **Website:** [NHTSA Intelligent Transportation](#)
- **Summary:** The NHTSA provides guidelines for traffic safety and infrastructure, including intelligent traffic systems. Their publications cover aspects of safety standards, traffic signal systems, and future developments in transportation technology.

Problem Statement

- **Problem 1: Traffic Congestion at Intersections**
- **Problem Statement:**
Traditional traffic light systems operate on **fixed timers**, which do not adapt to real-time traffic conditions. This leads to **unnecessary waiting times**, increased fuel consumption, and higher pollution levels. There is a need for a **smart, adaptive traffic light system** that can **dynamically adjust signal durations**.
- **Problem 2: Delay in Emergency Vehicle Movement**
- **Problem Statement:**
Emergency vehicles such as ambulances often get stuck in traffic due to **manual traffic management** and the absence of a priority system. This results in **delayed medical assistance** and potential loss of lives. A system that can **detect approaching emergency vehicles using GPS** and adjust signals to provide them a **clear passage** is required.

METHODOLOGY Planned For Your WORK

1. Data Collection & Sensing

- IR Sensors Deployment
- GPS-Based Ambulance Detection

2. Processing & Decision Making

- ESP32 Microcontroller Processing

3. Traffic Signal Control

- The microcontroller controls the traffic lights based on sensor data

System Model

Input	Processing (ESP32)	Output
IR Sensors (Lane Traffic Data)	Reads vehicle density	Controls traffic lights dynamically
GPS Data (Ambulance Location)	Detects ambulance presence	Gives emergency green signal
Microcontroller (ESP32)	Runs signal timing adjustment algorithm	Sends real-time updates (if needed)
(Optional) Cloud Integration	Stores & processes traffic data (Future Use)	Sends notifications to users

Expected Results

1. Efficient Traffic Flow Management

- The system dynamically adjusts signal timings based on real-time lane density, reducing congestion.

2. Priority for Emergency Vehicles

- GPS-based detection of ambulances ensures that they get a green signal, reducing emergency response time.

3. Reduced Waiting Time at Intersections

- The use of IR sensors helps optimize signal timing, minimizing unnecessary delays for vehicles.

4. Improved Road Safety

- Automated signal control reduces human errors in traffic management, preventing accidents and rule violations.

5. Scalability and Adaptability

- The system can be easily modified to integrate AI-based traffic prediction, vehicle counting, or smart city applications.

6. Power Efficiency

- The system ensures that signals operate with minimal power consumption and can incorporate solar panels for sustainability.



Applications of Your Work

Applications of Smart Traffic Control Management System

- **Urban Traffic Optimization** :Dynamic signal control & adaptive timing reduce congestion, improving traffic flow.
- **Emergency Vehicle Priority** :Detects emergency vehicles & clears paths by adjusting signals, reducing response time.
- **Environmental Benefits** :Reduces fuel consumption and CO2 emissions through efficient traffic management.
- **Pedestrian Safety** :Detects pedestrians & adjusts signals for safe crossings, enhancing safety at intersections.
- **Vehicle-to-Infrastructure (V2I) Communication** :Enables vehicles to communicate with traffic signals for real-time adjustments and safety alerts.
- **Real-Time Traffic Monitoring** :Provides real-time traffic updates & alerts to drivers and authorities via apps and dashboards.

References

Web & Online Sources:

- Smart City Initiative, "IoT-Based Traffic Control System," [Online]. Available: www.smartcityinitiatives.com/traffic-control
- "GPS-Based Emergency Vehicle Detection for Traffic Light Control," *TechWorld Blog*, 2023. [Online]. Available: www.techworld.com/emergency-gps-traffic



